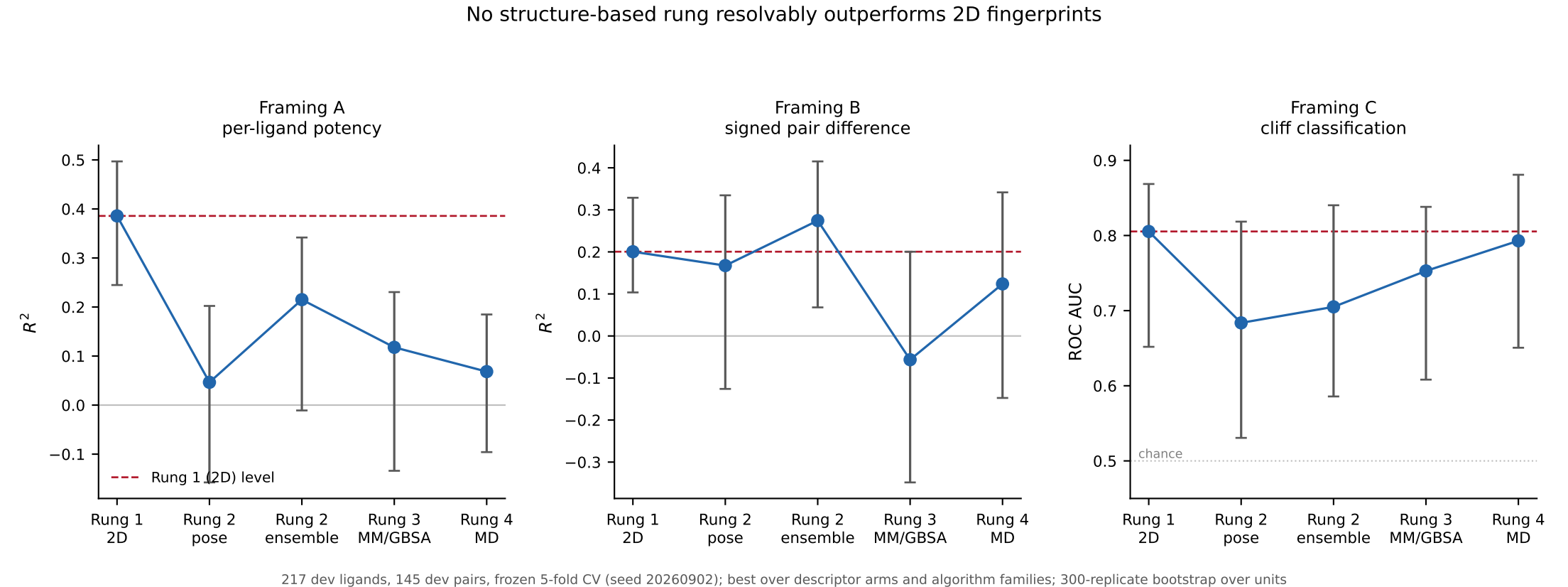
No structure-based rung resolvably outperforms 2D fingerprints
Framing A
per-ligand potency
R^2
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
Rung 1 (2D) level
Rung 1
2D
Rung 2
pose
Rung 2
ensemble
Rung 3
MM/GBSA
Rung 4
MD
Framing B
signed pair difference
R^2
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Rung 1
2D
Rung 2
pose
Rung 2
ensemble
Rung 3
MM/GBSA
Rung 4
MD
Framing C
cliff classification
ROC AUC
0.9
0.8
0.7
0.6
0.5
chance
Rung 1
2D
Rung 2
pose
Rung 2
ensemble
Rung 3
MM/GBSA
Rung 4
MD
217 dev ligands, 145 dev pairs, frozen 5-fold CV (seed 20260902); best over descriptor arms and algorithm families; 300-replicate bootstrap over units